Transaction Management

Agenda

- **What is Transaction?**
- **ACID Properties**

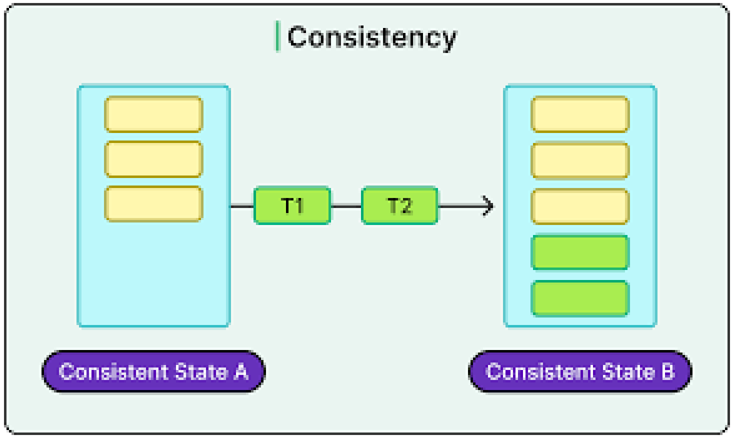
Transaction

Transaction

- **In DBMS, a transaction is a sequence of one or more database operations (such as INSERT, UPDATE, DELETE, or SELECT) that are executed as a single logical unit of work.**

Transaction

- **A transaction ensures that the database remains in a consistent state, even in the event of errors, system failures, or concurrent access.**



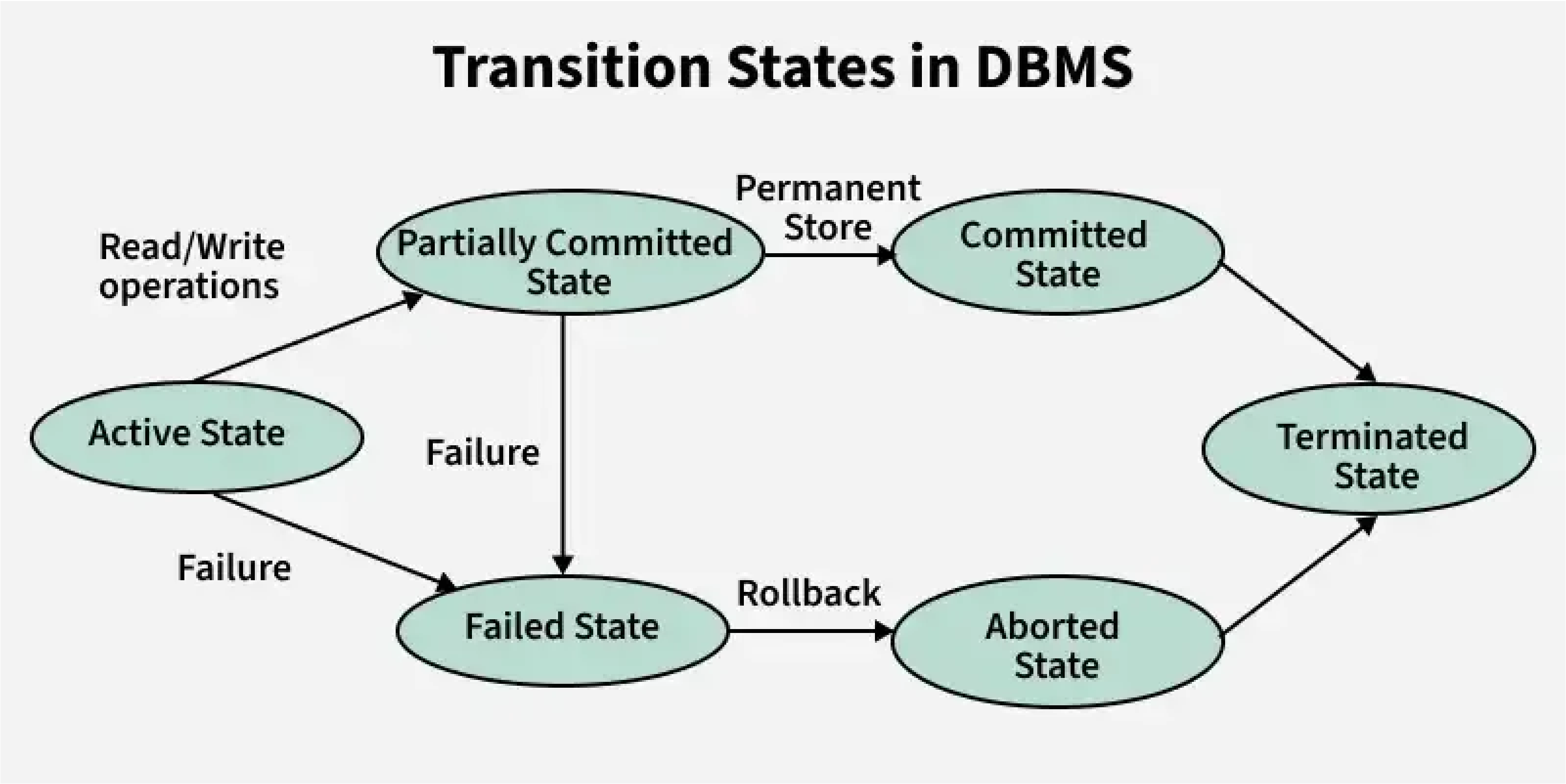
Transaction

- **After a failure, a DBMS ensures that the database is restored to a consistent state when the system restarts.**
- **A single execution of a user program may generate several transactions that perform specific database operations.**
- **Important characteristics of a transaction is that it consists of finite number of steps.**

Transaction - States

State	Description
Active	Transaction is executing operations.
Partially Committed	All operations done, waiting for commit.
Committed	Changes made permanent in the database.
Failed	Transaction cannot proceed due to an error or failure.
Aborted	All changes undone (rolled back) and transaction terminated.

Transaction States



Transaction - Operations

- **Read/Access data (R):** Accessing the database item from disk (where the database stored data) to memory variable.
- **Write/Change data (W):** Write the data item from the memory variable to the disk.
- **Commit:** Commit is a transaction control language that is used to permanently save the changes done in a transaction

Transaction - Sample

```
BEGIN TRANSACTION;

UPDATE Accounts SET Balance = Balance - 500 WHERE AccountID = 1;
UPDATE Accounts SET Balance = Balance + 500 WHERE AccountID = 2;

COMMIT;
```

Transaction - Sample

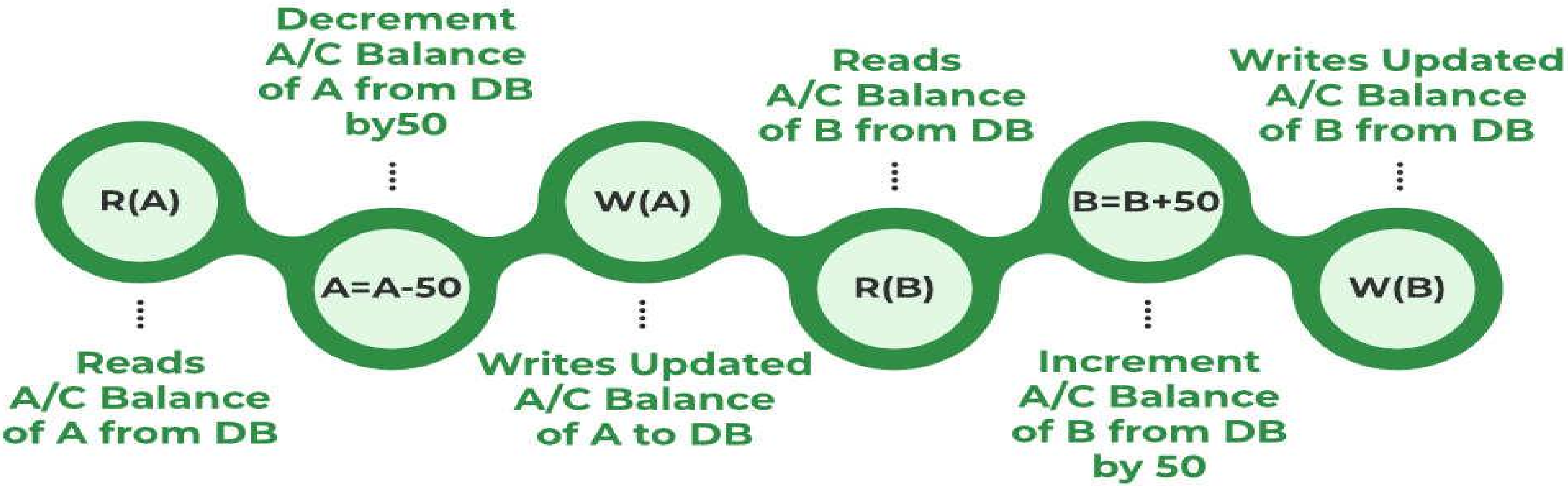
How many transactions?

UPDATE Accounts SET Balance = Balance - 500 WHERE AccountID = 1;

UPDATE Accounts SET Balance = Balance + 500 WHERE AccountID = 2;

Transaction - Example

- **Transfer of 50 PKR from Account A to Account B. Initially A= 500 PKR, B= 800 PKR. This data is brought to RAM from Hard Disk.**



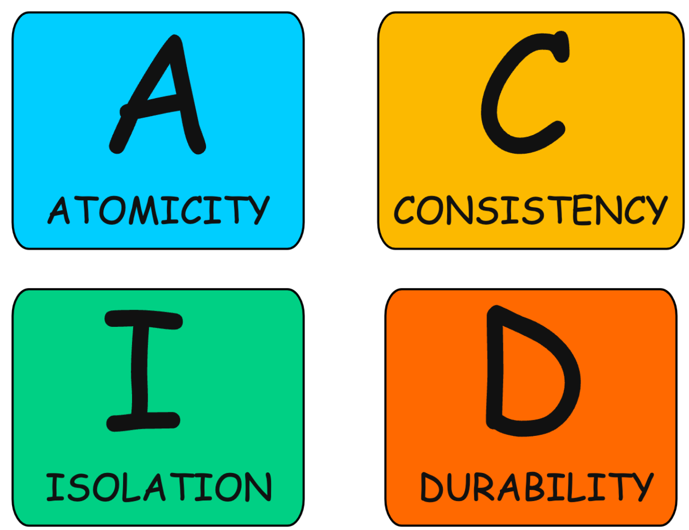
Transaction - Example

- **Note: The updated value of Account A = 450 PKR and Account B = 850 PKR.**
- **All instructions before committing come under a partially committed state and are stored in RAM. When the commit is read the data is fully accepted and is stored on a Hard Disk.**
- **If the transaction is failed anywhere before committing we have to go back and start from the beginning. We can't continue from the same state. This is known as Roll Back.**

ACID Properties

ACID Properties

- **A transaction in a Database Management System (DBMS) must follow four key properties known as ACID, which ensure reliable processing and maintain database integrity even in cases of errors or failures.**
- **A – Atomicity**
- **C – Consistency**
- **I - Isolation**
- **D - Durability**



ACID Properties - Atomicity

- **States that all operations of the transaction take place at once if not, the transactions are aborted.**
- **There is no midway, i.e., the transaction cannot occur partially. Each transaction is treated as one unit and either run to completion or is not executed at all.**

ACID Properties - Atomicity

- **Atomicity involves the following two operations:**
  - **Abort: If a transaction stops or fails, none of the changes it made will be saved or visible.**
  - **Commit: If a transaction completes successfully, all the changes it made will be saved and visible.**
- **Recovery Management Sub-system of DBMS support atomicity by using a variety of schemes, e.g. Shadow-database Scheme.**

ACID Properties - Consistency

- **The rules (integrity constraint) that keep the database accurate and consistent are followed before and after a transaction.**
- **When a transaction is completed, it leaves the database either as it was before or in a new stable state.**

ACID Properties - Consistency

- **This property means every transaction works with a reliable and consistent version of the database.**
- **The transaction is used to transform the database from one consistent state to another consistent state. A transaction changes the database from one consistent state to another consistent state.**

ACID Properties - Isolation

- **It shows that the data which is used at the time of execution of a transaction cannot be used by the second transaction until the first one is completed.**
- **In isolation, if the transaction T1 is being executed and using the data item X, then that data item can't be accessed by any other transaction T2 until the transaction T1ends.**
- **The concurrency control subsystem of the DBMS enforced the isolation property**

ACID Properties - Durability

- **Durability ensures that once a transaction has been committed, its changes are permanent, even if the system crashes or loses power immediately afterward.**
- **The recovery subsystem of the DBMS has the responsibility of Durability property.**

ACID Properties - Durability

- **How it works**

  - **DBMS uses logs (Write-Ahead Logging) and checkpoints.**
  - **Before committing, changes are recorded in a log file on non-volatile storage (like a hard drive).**
  - **On recovery, the DBMS replays or redoes committed changes if needed.**

THANK YOU